

CityWatch – Wearable Device and App-Based Emergency Response System

Sakshi Gangwani

Department of Computer Engineering
SVKM's Dwarkadas Jivanlal Sanghvi
College of Engineering
Mumbai, India
gangwani.sakshi15@gmail.com

Dhruvi Tank

Department of Computer Engineering
SVKM's Dwarkadas Jivanlal Sanghvi
College of Engineering
Mumbai, India
dhruvitank2403@gmail.com

Aksh Nishar

Department of Computer Engineering
SVKM's Dwarkadas Jivanlal Sanghvi
College of Engineering
Mumbai, India
akshnishar18@gmail.com

Prayag Mitaliya

Department of Computer Engineering
SVKM's Dwarkadas Jivanlal Sanghvi
College of Engineering
Mumbai, India
prayagmitaliya@gmail.com

Dr. Nilesh Patil

Department of Computer Engineering
SVKM's Dwarkadas Jivanlal Sanghvi
College of Engineering
Mumbai, India
nilesh.p@djsce.ac.in

Abstract— Emergencies such as harassment, assault, and unexpected medical conditions where the victim cannot manually call for assistance. Current alert system relies on manual intervention, which might not be possible at critical times. This paper introduces CityWatch, a wearable emergency system that automatically detects distress through machine learning and biometric sensors. Integrated with a web-based dashboard and real-time mobile application, CityWatch provides proactive safety through constant monitoring and instant alert dissemination. The system fills the gap between user-dependent tools and autonomous intervention solutions, providing a strong framework for personal safety in cities or villages. Once detected, the system automatically notifies emergency contacts and officials, sending live location details. The system also has a proactive safety feature that warns users when they are entering a high-crime zone. The system further automatically escalates the alert to higher officials for urgent action if officials do not act on a case within a specified period of time. A mobile app enables users to report emergencies manually, upload photos or video messages, and monitors case status. Emergency services get detailed reports through a linked dashboard, where they can view the situation in real-time.

Keywords—Emergency response, wearable technology, IoT, machine learning, safety system, crime prevention, high-crime area alerts, authority escalation, location tracking, real-time reporting, real-time monitoring.

I. INTRODUCTION

Both rural and urban environments are increasingly confronted with all types of harassment, assault, medical emergencies, and other emergencies requiring immediate and effective action. Such emergencies' conventional reporting method is greatly reliant on the capacity of victims or witnesses to trigger an alarm manually, which is usually impossible or difficult at the decisive moment. For instance, an individual having a panic attack, a medical distress, or an attack may physically not be able to grab their phone or communicate with emergency systems in real time. Additionally, even where incidents are reported, there is frequently no immediate means for the victim to monitor the progress of the emergency response or confirm that it is being addressed as it should be. Therefore, such response time and awareness gaps can lead to dramatic results, particularly in cases where the victim is himself unable to perform necessary acts.

To address these challenges, the need for a more autonomous, real-time, and proactive solution to personal safety has never been more critical. CityWatch is an emergency response wearable system that combines real-time physiological monitoring, intelligent distress detection, and location-based alerting to overcome the limitations of existing emergency reporting frameworks. Building on the assistance of cutting-edge technologies such as machine learning, wearable sensors, and cloud communication, CityWatch can autonomously detect signs of distress, such as irregular heart rates or unusual movements, and automatically trigger alerts to emergency contacts, authorities, and concerned services.

One of the distinguishing qualities of CityWatch is that it sends messages automatically when users enter crime zones, an anticipatory safety system that minimizes danger before it becomes a threat. This is especially useful in urban settings where danger is more likely to be encountered. CityWatch also provides smart escalation of notifications. If the authorities fail to act on an alert within a reasonable period, the system will escalate the case automatically to higher authorities so that urgent cases are not overlooked or disregarded.

The system is also made extremely user-friendly, with an app that allows users to manually report emergencies, track the status of ongoing cases, and transmit images or voice messages to help the response process. This has the added benefit of enabling users to engage with the system without manually dialing emergency numbers. The supporting emergency responder dashboard enables them to obtain real-time information, with all case details, user details, and response time recorded for purposes of transparency and accountability.

This paper presents the design, development, and implementation of *CityWatch*, detailing its core features, the underlying technology stack, and its ability to autonomously detect and respond to emergencies. The system aims to enhance the safety and security of individuals in both rural and urban environments by ensuring rapid response times, improving transparency in emergency management, and ultimately saving lives.

The paper is organized as follows. Section II presents related work, presenting similar systems and research papers in real-time emergency response systems. Section III presents the

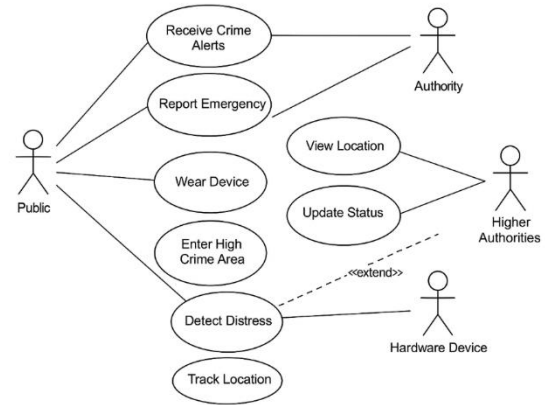
user groups and features implemented in the proposed system. Section IV presents the system architecture and components. Section V presents the implementation of the CityWatch system, including the design and development phase. Section VI presents the results and performance analysis of the system. Section VII concludes the paper.

II. EXISTING SYSTEMS

The demand for increased personal security, especially among vulnerable groups such as women, in urban and rural settings has driven extensive research into technology-based emergency response systems [1, 13]. Conventional methods based on manual notification are usually insufficient in life-threatening situations where a victim is incapacitated [1, 15]. Wearable technology provides a fundamental solution, with built-in sensors like pulse-rate, temperature, accelerometers, and gyroscopes measuring the physiological status of the wearer and flagging anomalies consistent with distress, for example, falls or sudden onset of abnormal vital signs [1, 2, 16, 17]. Examples of such systems designed by Hyndavi et al. [1] and Lall et al. [14] are incorporating sensors in bands or other wearable forms for permanent monitoring. The information collected by these Internet of Things (IoT) based wearables [4, 13] needs to be processed with Machine Learning (ML) or Deep Learning (DL) algorithms in many cases to effectively differentiate between real emergencies and normal daily activities and minimize false alarms [5, 16]. Such AI-based analysis is important to facilitate automatic generation of alerts without the need for direct user intervention [5, 8, 13]. Location-based services (LBS), drawing largely on GPS, are central to identifying the location of the victim [1, 6, 7, 11] and supporting geographically sensitive responses. In addition, LBS offers proactive safety features, for instance, alerts for users upon approaching pre-specified areas with high crime rates, a feature discussed in campus safety applications such as WalkSafe [7]. Communication protocols are crucial for relaying alerts; although cellular (GSM/GPS) is prevalent [1, 11], substitutes such as LoRa are being explored for situations with inadequate conventional network coverage [6]. Mobile applications are the major interface for users, enabling users to manually request SOS alerts, see safety data (such as crime hotspots based on GIS analysis [11]), monitor the status of an alert, and configure emergency contacts [7, 9, 11, 12]. Systems are trending towards more unified models, incorporating crime reporting, analysis, response, and prevention and regularly building on society participation in which local volunteers or the public can be notified in addition to official responders [9, 10, 11]. Platforms such as those presented by Bhargavi et al. [9] and Mahmoud et al. [10] are aimed at simplifying the reporting and management process for citizens and authorities. The grand idea of an Internet of Emergency Services (IoES) attempts to integrate these disparate technologies—wearables, IoT sensors, AI/ML, LBS, mobile apps, communication networks, and backend platforms—into a harmonious and responsive ecosystem [4]. CityWatch elaborates on such developments by integrating automated distress triggering through wearables, proactive location-based alerting, AI-based analysis, easy-to-use mobile interaction, and smart alert escalation, which tackles significant limitations of manual

trigger and disjointed response found in existing work [13, 15].

III. PROPOSED SYSTEM



1. Public (End Users):

The general public are the primary users of the CityWatch system. They wear the hardware device, which automatically records biometric information (e.g., heart rate, movement) to detect distress signals automatically. In case of an emergency, users can report manually, and incidents will be included along with location, images and video. The system also warns users when they approach locations with a high crime risk, keeping them away from potential threats.

2. Authority (Emergency Responders):

Emergency responders such as police, medical, and fire services are notified in real-time upon the activation of an alert by the user. They are offered real-time location and distress data of the user via an easy-to-use dashboard. This enables them to quickly evaluate the situation and set priorities for the response. The dashboard also features updating the status of cases, enhancing coordination and timely interventions.

3. Hardware Device (Wearable):

The wearable device is at the heart of the CityWatch system, continuously tracking the biometric data of the user. The unit is also equipped with sensors that track abnormal physiological patterns that would be indicative of an emergency, such as an abnormal spike in heart rate or abnormal movement. The unit also tracks the GPS location of the user for real-time location tracking during an emergency. The unit is also tasked with alerting the user and emergency responders during distress.

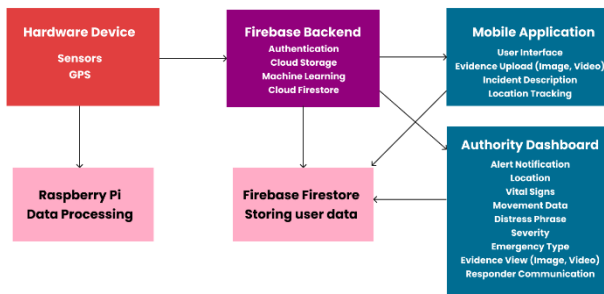
4. Higher Authorities (Supervisory Authorities):

Higher authorities also have the responsibility of overseeing the overall performance of the system. They will receive incoming emergency reports and responses. This allows them to monitor the progress of current cases, track response times, and make sure that emergencies are given the importance

they deserve. In case a local authority does not respond within a reasonable time, the system automatically escalates the case to higher authorities so that there can be accountability and no case of importance goes unnoticed.

IV. SYSTEM ARCHITECTURE

The architecture of the CityWatch system can be broken down into three key components:



System Architecture of CityWatch

1. Hardware (Wearable Device)

The wearable device itself is the system's point of focus. It constantly monitors the user's array of biometric and environmental data. It has:

Sensors:

Accelerometer & Gyroscope: Monitors movement patterns, acceleration, and orientation. Employed to detect unusual activity, falls, or distress calls.

Heart Rate Sensor: Tracks the user's heart rate to detect stress, anxiety or physical distress.

GPS Module: Tracks the true user location in real-time to provide accurate positioning to emergency services and crime hotspot mapping.

Processing Unit:

The Raspberry Pi can act as the processing unit of the wearable device, communicating with the various sensors (heart rate & SpO2 sensor, microphone, accelerometer, etc.) and sending the information to the firebase. It will handle the interaction between the sensors and Firebase, sending data like heart rate, accelerometer readings, microphone input, and GPS coordinates to the cloud in real-time.

2. Mobile Application

The mobile application integrates the wearable device to Firebase and provides the user interface. Firebase Authentication is used by it to facilitate secure sign-in and identity management.

Authentication:

They can log in with their credentials (e.g., Google Sign-In, Email/Password) through Firebase Authentication.

Real-Time Data:

Firebase Database is used to store and retrieve the biometric data, location, and emergency notifications. When the

wearable device collects data, the application updates the Firebase database periodically, which is real-time synchronized to the application.

Emergency Alerts:

When the app detects unusual readings from the wearable or when an event is reported manually by a user, Firebase triggers an alert.

Crime Hotspot Identification:

Firebase stores the crime hotspot data in Firestore. The app requests Firebase provide it with nearby hotspot locations around the user's current location and notifies them if they are near a high-risk area.

3. Backend System with Firebase:

Since Firebase is a backend, it simplifies the architecture by taking care of the storage, real-time synchronization, and authentication without the need to install a traditional server.

Firebase Services Utilized:

Cloud Firestore: For storing user health real-time data, emergency reports, crime hotspots, and location tracking.

Firebase Database: For real-time updating and synchronization of sensor data (heart rate, movement patterns, location).

Firebase Authentication: Safe sign-in for users and identity management.

Firebase Cloud Messaging (FCM): For sending notifications to users or authorities regarding emergency incidents or proximity to crime hotspots.

V. IMPLEMENTATION

Implementation of the CityWatch system integrates a synergistic combination of hardware, software, machine learning, and cloud-based technologies to provide real-time, autonomous emergency response. In a bid to operate smoothly across various user roles such as victims, witnesses, responders, and supervisory authorities, the system consists of five fundamental building blocks: a cross-platform mobile app for users, an emergency services web dashboard, a wearable hardware component for physiological monitoring in real-time, intelligent machine learning models for automated detection of distress, and a single backend infrastructure with Firebase as the power source. Each of the components has been designed with utmost care to be responsive, accurate, and usable with low latency and high availability in both urban and rural settings. The subsequent sections give a detailed overview of the design and integration of these components.

A. Mobile Application – CityWatch Victim & Witness App

The mobile application which was developed as part of the CityWatch system is primarily for victims and witnesses. It offers a simple and user-friendly interface for reporting emergencies. It is developed using React Native, which offers cross-platform compatibility for Android and iOS devices.

The users need to authenticate themselves with Firebase Authentication such as email-password or Google sign-in. After logging into the app, users can report incidents by choosing the nature of the emergency themselves or by attaching supporting evidence in image and video formats. The app also records the GPS location of the user and reports the incident in real time to the cloud backend on Firebase.

Additionally, the app allows ongoing tracking of reported cases, in which users can see the progress of their reports, marked as active, solved, or ignored. One of the key features of the application is its crime hotspot notification. By pulling location data from Firestore, the app can alert users when they approach areas that have been marked as high-risk. This situational awareness mechanism provides real-time alerts and promotes safer decision-making. Firebase Cloud Messaging-powered push notifications alert users to updates on their cases or other emergencies in the vicinity, encouraging personal safety and neighborhood watch.

B. CityWatch Dashboard – Emergency Response Units

CityWatch's web-based dashboard is particularly created for emergency response officials like police officers, paramedics, and fire department officials. Created with React, the dashboard provides secure login using Firebase Authentication and role-based permissions for responders and supervisors at a higher level. Upon authentication, officials are provided with a categorized listing of all reported cases—active, closed, and ignored—and detailed data on each case, including the time of incident, user's vitals, media evidence, and precise location coordinates.

Its capabilities include updating case statuses, internal task assignment, or pushing unresolved alerts to higher levels of government. Synchronization between the dashboard and Firebase Firestore in real-time ensures updates by responders are transferred immediately across the platform, including the victim's app. The dashboard also offers the ability to manually add new cases, which is beneficial during patrols or citizen-to-citizen contact. In all, the dashboard serves as the command center for incidents at a city-wide level, ensuring accountability and rapid decision-making by response units.

C. Hardware – Wearable Device Prototype

The wearable device prototype created for CityWatch is the foundation of its independent emergency detection feature. Constructed around a Raspberry Pi module, the gadget incorporates several biometric and motion sensors that monitor the physical state of the wearer continuously. They include a sensor for accelerometer and gyroscopic data, a sensor for heart rate and SpO₂ levels, a temperature sensor for body heat monitoring, and a microphone for sound input. There is also an embedded emergency button for alerts by hand during critical moments. Combined, these elements allow the device to identify falls, increases in physiological stress, or distress voice commands without involvement from the user.

Sensor data is interpreted using Python scripts on the Raspberry Pi. The scripts use multithreading to read and send

sensor data to Firebase in real-time simultaneously. When the readings exceed pre-set emergency levels—such as a great fall impact, low SpO₂, or distress words—the device automatically raises an alarm in Firebase, which is then sent to the respective authorities via the mobile and dashboard systems. There is also a built-in GPS module that ensures proper location monitoring, facilitating prompt intervention by the emergency services.

D. Machine Learning Services

CityWatch uses three separate machine learning models to improve its smart emergency detection feature. The models run either on the wearable device for edge processing or on cloud-based Firebase functions, based on computational intensity and the urgency of analysis. The first is a fall detection model that analyzes accelerometer and gyroscope data to detect abrupt changes in orientation and speed, which could indicate a physical fall. A decision tree classifier that is trained on actual movement data guarantees fall detection reliability.

The second model targets stress detection by examining inputs like heart rate, SpO₂, temperature, and small movement patterns. Executed using LightGBM, this model classifies the user's level of stress into normal, moderate, or high, enabling the system to schedule alerts accordingly. The third model hears microphone input for the identification of pre-defined help keywords in English as well as Hindi such as "help," "emergency," "save me", "bachao", "madad karo", etc. These ML services are the basis for CityWatch's autonomous alert system, reducing false positives and accurate identification of critical situations.

E. Database – Firebase Firestore and Related Services

The CityWatch system uses Google Firebase as its integrated cloud backend, allowing smooth communication and synchronization among all its parts. Cloud Firestore is used as the main database for storing structured data such as user profiles, case reports, media file URLs, location coordinates, and health metrics. The Firestore database schema is designed for quick querying and real-time updates, making both the mobile app and the dashboard synchronized with the latest data.

Supplementing Firestore, the Firebase Realtime Database is utilized for real-time streaming of sensor data with high frequencies, particularly those collected from the wearable device. Firebase Cloud Storage manages uploading and fetching multimedia content, including images and audio evidence that users provide. Firebase Cloud Messaging also provides prompt delivery of reminders and notifications to authorities and users. User authentication throughout the system is handled by Firebase Authentication, which provides several secure login modes. Collectively, the services provide a solid, scalable, and effective backend environment that supports CityWatch's real-time safety system.

VI. RESULTS

The hardware component of the CityWatch system – the wearable device & its functionalities are depicted in the following figures.

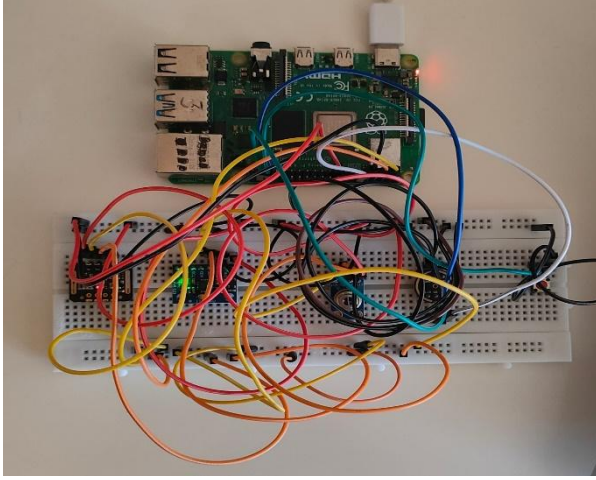


Fig. 1. Wearable Device - Prototype

```
[ALERT] Help Keyword detected multiple times. Alerting authorities!

[INFO] Data to be written to Firestore:
{
  "description": "Autodetected by the device",
  "location": {
    "latitude": 19.0728,
    "longitude": 72.8826
  },
  "address": "SVKM's Dwarkadas J. Sanghvi College of Engineering, Vile Parle, Maharashtra, India, 400056",
  "photo": [],
  "type_of_event": "Wearable Device",
  "user_id": "VjoXUDwAI7Q0FxfRdOmGagX0Yk1",
  "user_type": "Victim",
  "video": [],
  "authority_id": 1,
  "body_temp": 31.010000000000048,
  "heart_rate": 77.25,
  "blood_oxygen": 99.657,
  "id": "6404f51e-7739-4ffd-8d15-7235ecb44002",
  "date_time": "2025-04-13T07:55:44.689528+00:00"
}
```

Figure 2. One of the help keywords has been detected more than once, which indicates that someone is in a bad situation and requires help. The system will automatically notify the right authorities so they can arrive as quickly as possible.

```
[ALERT] Help Keyword and High Stress detected. Alerting authorities!

[INFO] Data to be written to Firestore:
{
  "description": "Autodetected by the device",
  "location": {
    "latitude": 19.0728,
    "longitude": 72.8826
  },
  "address": "SVKM's Dwarkadas J. Sanghvi College of Engineering, Vile Parle, Maharashtra, India, 400056",
  "photo": [],
  "type_of_event": "Wearable Device",
  "user_id": "VjoXUDwAI7Q0FxfRdOmGagX0Yk1",
  "user_type": "Victim",
  "video": [],
  "authority_id": 1,
  "body_temp": 38.870000000000005,
  "heart_rate": 96.25,
  "blood_oxygen": 98.687,
  "id": "ba46f2c4-7f06-41ea-883a-3baecd9a9c6f",
  "date_time": "2025-04-13T07:53:24.481457+00:00"
}
```

Figure 3. One of the help keywords has been detected along with high stress level. It indicates that an emergency has arrived. The system notifies the relevant authorities so that they can take an action quickly.

```
[ALERT] Fall and Help Keyword detected. Alerting authorities!

[INFO] Data to be written to Firestore:
{
  "description": "Autodetected by the device",
  "location": {
    "latitude": 19.0728,
    "longitude": 72.8826
  },
  "address": "SVKM's Dwarkadas J. Sanghvi College of Engineering, Vile Parle, Maharashtra, India, 400056",
  "photo": [],
  "type_of_event": "Wearable Device",
  "user_id": "VjoXUDwAI7Q0FxfRdOmGagX0Yk1",
  "user_type": "Victim",
  "video": [],
  "authority_id": 1,
  "body_temp": 31.550000000000001,
  "heart_rate": 64.75,
  "blood_oxygen": 95.70065999999999,
  "id": "46d8f744-2d3f-43b0-881e-d919d92fe5c6",
  "date_time": "2025-04-13T08:13:43.638635+00:00"
}
```

Figure 4. A fall has been detected together with one of the help keywords, suggesting that someone might be injured or in danger. As a result, an emergency alert is sent to notify the authorities for a rapid response.

```
[ALERT] High Stress and Fall detected. Alerting authorities!

[INFO] Data to be written to Firestore:
{
  "description": "Autodetected by the device",
  "location": {
    "latitude": 19.0728,
    "longitude": 72.8826
  },
  "address": "SVKM's Dwarkadas J. Sanghvi College of Engineering, Vile Parle, Maharashtra, India, 400056",
  "photo": [],
  "type_of_event": "Wearable Device",
  "user_id": "VjoXUDwAI7Q0FxfRdOmGagX0Yk1",
  "user_type": "Victim",
  "video": [],
  "authority_id": 2,
  "body_temp": 31.410000000000025,
  "heart_rate": 125.5,
  "blood_oxygen": 99.796266,
  "id": "455a2e67-0a50-4733-83e1-2691ed1eed9c",
  "date_time": "2025-04-13T08:35:25.989399+00:00"
}
```

Figure 5. A combination of high stress and a fall is detected indicating an emergency. To ensure timely response an alert is sent to the appropriate authorities.

```
[ALERT] Emergency button is pressed. Alerting authorities!

[INFO] Data to be written to Firestore:
{
  "description": "Detected by the device",
  "location": {
    "latitude": 19.0728,
    "longitude": 72.8826
  },
  "address": "SVKM's Dwarkadas J. Sanghvi College of Engineering, Vile Parle, Maharashtra, India, 400056",
  "photo": [],
  "type_of_event": "Wearable Device",
  "user_id": "VjoXUDwAI7Q0FxfRdOmGagX0Yk1",
  "user_type": "Victim",
  "video": [],
  "authority_id": 2,
  "body_temp": 31.290000000000002,
  "heart_rate": 115.25,
  "blood_oxygen": 98.169384,
  "id": "3646dcab-1a00-40a7-a874-3a9c255cch9e",
  "date_time": "2025-04-13T08:29:21.528629+00:00"
}
```

Figure 6. Emergency button is pressed indicating a critical situation. An alert is sent to the authorities for effective response.

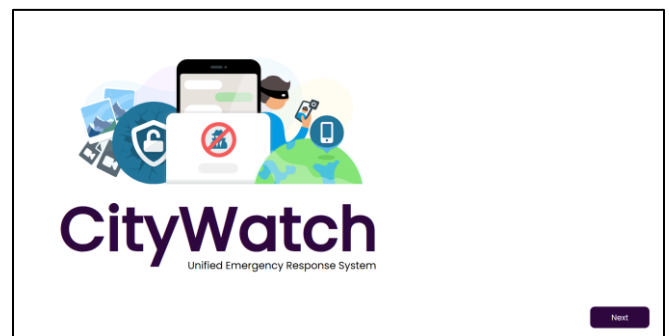


Figure 7. CityWatch Dashboard Home Page

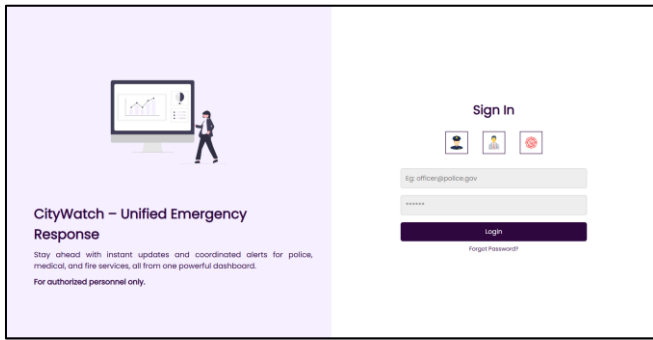


Figure 8. CityWatch Dashboard Authentication

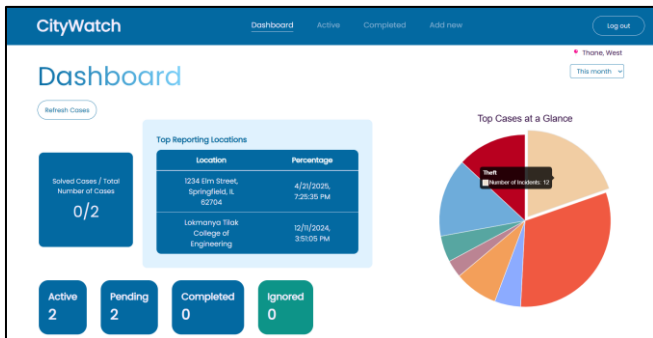


Figure 9. CityWatch Dashboard

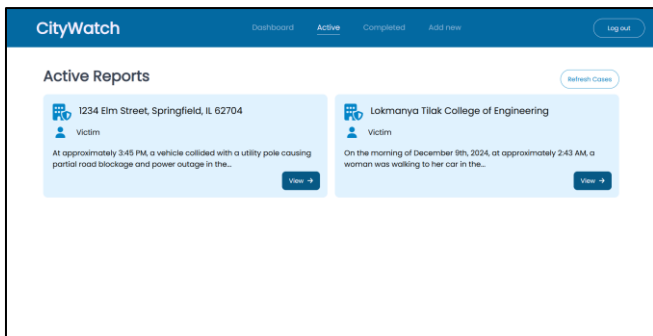


Figure 10. CityWatch Dashboard Active Cases

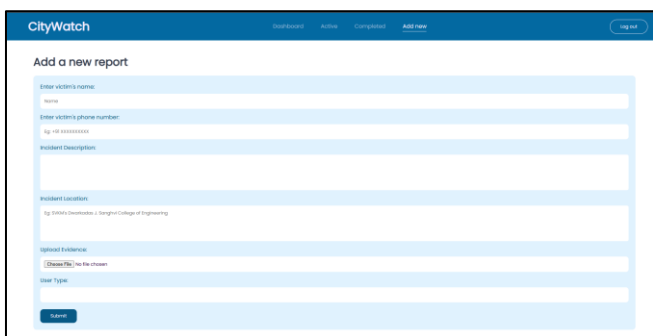


Figure 11. CityWatch Dashboard Add New Case

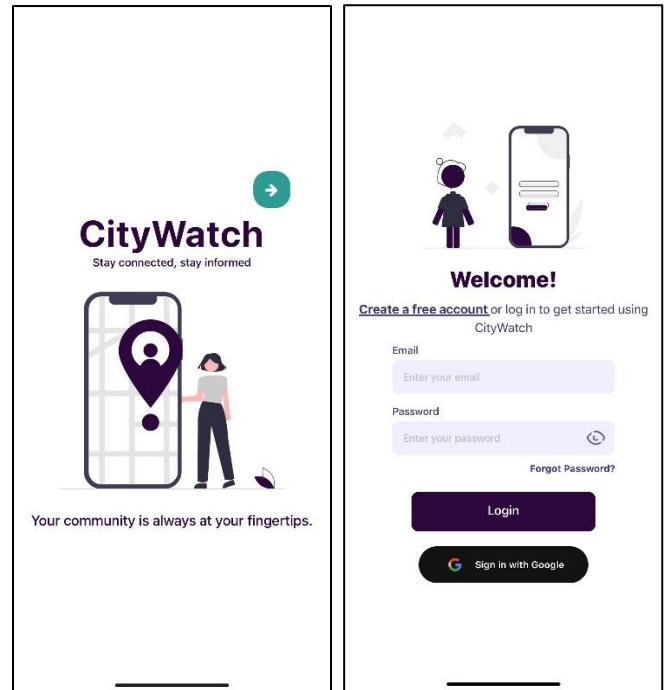


Figure 12. CityWatch App Authentication

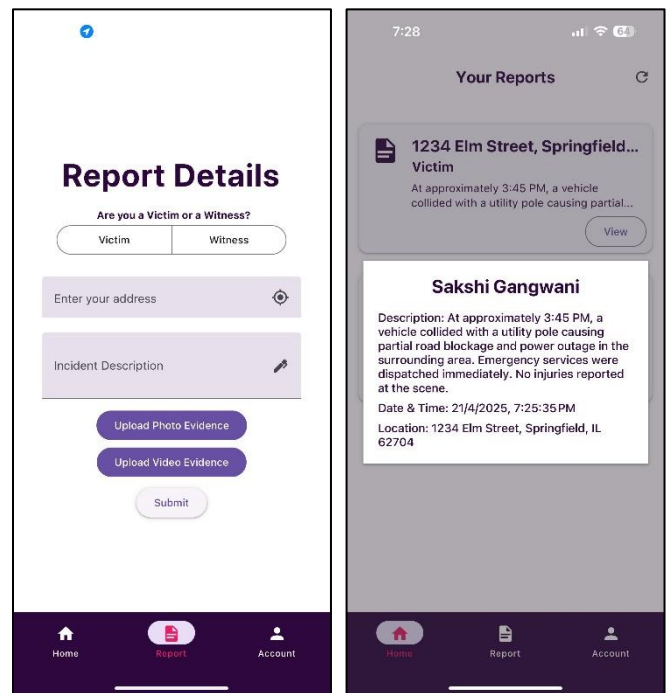


Figure 13. CityWatch App Case Reporting and Details

VII. CONCLUSION

CityWatch is a major advance in personal protection through automatic activation of alarm messages based on physiological information, independent of manual initiation. CityWatch leverages real-time monitoring of vital signs to identify distress and alert authorities in real-time, which leads to quicker response and possibly saves lives. CityWatch finally provides a more robust, independent solution to personal protection, opening the way for more intelligent, quicker responses to emergency.

VIII. FUTURE SCOPE

To raise the contextual awareness of the CityWatch system, future releases can include sensors like electrodermal activity (EDA) for more accurate identification of stress levels, as well as environmental audio sensors that can detect emergency-related acoustic signals like shouting, loud crashes, or breaking glass. Including sensors for ambient temperature and air quality can also increase the system's ability to respond to certain environmental emergencies. The integration of smart data processing methods will enable the system to process these varied inputs together, leading to better emergency detection and less false alarms. Beyond detection, predictive modeling can be investigated to predict risk levels from historical and real-time data from sources such as user reports, local news APIs, or public social media streams. Adding on a smart navigation feature that recommends safer routes based on dynamic risk patterns would further turn CityWatch into a proactive personal security buddy.

REFERENCES

- [1] Hyndavi, V., Sai Nikhita, N., & Rakesh, S. (2020, July). Smart Wearable Device for Women Safety Using IoT. In 2020 5th International Conference on Communication and Electronics Systems (ICCES) (pp. 459-463). IEEE.
- [2] Mohandes, M., Rojas, J. P., Rehman, M. U., Hussein, A., E'mar, A., AlRaei, Y., & AlZaher, H. (2022). Kirigami-Enabled Wearable Health and Crowd Monitoring System. *Arabian Journal for Science and Engineering*, 47(3), 3583-3595.
- [3] Tanwar, R., Nandal, N., Zamani, M., & Manaf, A. A. (2022). Pathway of Trends and Technologies in Fall Detection: A Systematic Review. *Healthcare*, 10(1), 172.
- [4] Damaševičius, R., Bacanin, N., & Misra, S. (2023). From Sensors to Safety: Internet of Emergency Services (IoES) for Emergency Response and Disaster Management. *Journal of Sensor and Actuator Networks*, 12(3), 41.
- [5] Gowda, D. V., Damacharla, P., Maddineni, V. K., & Gummadi, V. A. K. (2024, March). Deep Learning Approaches for Real-Time IoT Data Processing and Analysis. In 2024 5th International Conference on Smart Electronics and Communication (ICOSEC) (pp. 757-764). IEEE.
- [6] Sciallo, L., Trotta, A., & Di Felice, M. (2020). Design and performance evaluation of a LoRa-based mobile emergency management system (LOCATE). *Ad Hoc Networks*, 96, 101993.
- [7] Vaghela, S. J. D. (2015). Walksafe: College campus safety app incorporating pebble smartwatch (Master's thesis, The Pennsylvania State University).
- [8] Reddy, M. S., Vamsi, C., & Kathambari, P. (2024, March). Rescue me: AI Emergency Response and Disaster Management System. In 2024 2nd International Conference on Artificial Intelligence and Machine Learning Applications (AIMLA) (pp. 1-5). IEEE.
- [9] Bhargavi, G., Vaishnavi, & Chopra, P. (2024, May). Automated Crime Reporting and Incident Management System. In 2024 International Conference on System, Computation, Automation and Networking (ICSCAN) (pp. 1-5). IEEE.
- [10] Mahmoud, A. A. S., Mandela, N., Agrawal, A. K., & Mistry, N. R. (2023, November). Online Crime Reporting System for Digital Forensics. In 2023 International Conference on Quantum Technologies, Communications, Computing, Hardware and Embedded Systems Security (iQ-CCHES) (pp. 1-6). IEEE.
- [11] Shenoy, M. V., Sridhar, S., Salaka, G., Gupta, A., & Gupta, R. (2021). A Holistic Framework for Crime Prevention, Response, and Analysis With Emphasis on Women Safety Using Technology and Societal Participation. *IEEE Access*, 9, 66188-66207.
- [12] Abidi, S. I. A., Almeida, A. A., Soares, L. G., & Pansare, A. (2021, December). Interactive Map Application For Real-Time Crime Reporting. In 2021 International Conference on Advances in Computing, Communication, and Control (ICAC3) (pp. 1-8). IEEE.
- [13] Farooq, M. S., Masooma, A., Omer, U., Tehseen, R., Gilani, S. A. M., & Atal, Z. (2023). The Role of IoT in Woman's Safety: A Systematic Literature Review. *IEEE Access*, 11, 69807-69825.
- [14] Mahmoud, A. A. S., Mandela, N., Agrawal, A. K., & Mistry, N. R. (2023, November). Online Crime Reporting System for Digital Forensics. In 2023 International Conference on Quantum Technologies, Communications, Computing, Hardware and Embedded Systems Security (iQ-CCHES) (pp. 1-6). IEEE. (Note: Paper [10] and [15] are the same reference, cited twice in the paragraph).
- [15] Lall, P., Zhang, H., & Rahul, R. (2018, May). Flexible wearable biometric band and smartphone application for prevention of sudden causes of death. In 2018 IEEE 68th Electronic Components and Technology Conference (ECTC) (pp. 1790-1797). IEEE.